Weighted doubling: local field preservation

z,w,y,2 nonzero + normalized point equation

Normalize literal output → $u=(X-xx)^2$, $v=sX+(yy-s\cdot xx)$

$s=f'(xx)/(2yy)$; explicit Taylor quotient → u divides $f-v^2$

y=0 uses earlier inverse branch. Zero derivative is permitted.